Charge transfer in core-shell nanostructures

"How does shell order affect charge transfer in SiO2/MoS2/ZnS structures?" *Generated by LeapSpace*

Shell order changes charge transfer by deciding which layer sits at the key interface: a MoS_2-outer design should favor electron capture/transport at the surface, while a ZnS-outer design should more strongly passivate and shift where carriers are injected, so the net transfer pathway is likely different even if the same three materials are present. [131] [132] [149] [150]

What the evidence implies for SiO2/MoS2/ZnS vs SiO2/ZnS/MoS2

Shell order	Expected charge-transfer effect	Evidence-based interpretation
SiO_2 core / MoS_2 inner shell / ZnS outer shell	Carriers generated in or near the outer ZnS layer must cross into MoS_2 only after passing the ZnS/MoS_2 interface; charge separation can still be efficient if the MoS_2 layer remains an energy bridge, but the outer ZnS can also act as a barrier/passivation layer [131] [149] [150]	This order is more "passivation-first": the outer ZnS can suppress recombination and protect the inner MoS_2, but it may slow direct access of carriers to MoS_2 unless the interface is highly coupled [131] [150]
SiO_2 core / ZnS inner shell / MoS_2 outer shell	Carriers reaching the surface can transfer directly into the outer MoS_2, which is repeatedly described as a favorable charge-transport and charge-capture layer [132] [151] [152]	This order is more "transport-first": ZnS can act as a scaffold/interlayer, while outer MoS_2 is better positioned to collect electrons and reduce interfacial recombination [132] [151] [152]

Why shell order matters

- **MoS_2 often acts as the charge-transport or energy-bridge layer.** In a Si/MoS_2/TiO_2 heterostructure, the MoS_2 layer is explicitly described as an energy bridge that reduces recombination at the direct Si/TiO_2 contact [151].
- **ZnS/MoS_2 interfaces are charge-separating interfaces.** A MoS_2/ZnS junction was reported to improve segregation of photogenerated carriers, with MoS_2 integrated onto ZnS stimulating sulfur vacancies and narrowing the band gap [132].
- **Interlayer order changes charge-transfer direction and magnitude in layered systems.** Across multiple heterostructures, stacking order, interface structure, and interfacial coupling strongly alter charge redistribution and transfer rates [153] [154] [155] [156].

- **A continuous intermediate layer can preserve shell growth and improve coupling.** In $ZnS/Al_2O_3/MoS_2$ nanowires, the Al_2O_3 interlayer preserved a smooth surface for continuous MoS_2 shell growth, showing that the identity and sequence of layers directly affect interface quality and therefore charge transport [131].
- **Shell sequence controls whether the outer surface is a transport layer or a protective layer.** ZnS-based heterojunctions can enhance separation by donating electrons to a partner layer, while MoS_2-rich outer surfaces are repeatedly associated with stronger charge capture and transport [132] [150] [157].

Practical takeaway

If your goal is **maximizing charge transfer to the outside**, the **$SiO_2/ZnS/MoS_2$** order is the more favorable design logic because MoS_2 is placed where it can act as the outer charge-collecting shell [132] [151] [152]. If your goal is **protecting the structure and suppressing recombination**, **$SiO_2/MoS_2/ZnS$** can still work, but it is more likely to behave as a passivated architecture where ZnS moderates access to the MoS_2 layer [131] [150]. That is an inference from adjacent heterostructures, not a directly measured $SiO_2/MoS_2/ZnS$ result [131] [132].

Bottom line: shell order should be chosen by function—**MoS_2 outside for easier charge extraction, ZnS outside for stronger passivation and protection**—because the evidence consistently shows that interfacial sequence, not just composition, governs charge transfer pathways. [131] [132] [150] [151] [153]

Confidence level: **Medium** — the direction is strongly supported by closely related heterostructures, but the exact $SiO_2/MoS_2/ZnS$ stack has not been directly benchmarked in the retrieved evidence.

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